PYTHON ACTIVITY BASED LEARNING

PROBLEM –1:

Student Feedback Sentiment Scoring: Analyse multiple feedback sentences using a supplied keyword list, count positive and negative indicators, calculate scores, and classify feedback using predefined thresholds.

Objective:

To analyze student feedback by identifying positive and negative keywords, calculating a sentiment score, and classifying the feedback as Positive, Negative, or Neutral based on user input.

Algorithm

1. Start.
2. Initialize the lists of positive words and negative words.
3. Read the number of feedback sentences, n, from the user.
4. Repeat steps 5–12 for each feedback sentence from 1 to n.
5. Read a feedback sentence from the user.
6. Convert the sentence to lowercase and remove common punctuation.
7. Split the sentence into individual words.
8. Initialize positive_count = 0 and negative_count = 0.
9. For each word in the sentence:
    - If the word is in the positive words list, increment positive_count.
    - If the word is in the negative words list, increment negative_count.
10. Calculate the sentiment score:

score = positive_count - negative_count

11. Classify the sentiment:

- If score > 0, set sentiment to Positive.

- If score < 0, set sentiment to Negative.
- Otherwise, set sentiment to Neutral.

12.Display the feedback, positive indicators, negative indicators, sentiment score, and classification.

13.Stop.

Pseudo code:

```
BEGIN
    positive_words = ["good", "excellent", "helpful",
                      "clear", "interesting"]
    negative_words = ["bad", "poor", "boring",
                      "difficult", "confusing"]
    INPUT n
    FOR i = 1 TO n DO
        INPUT feedback
        CONVERT feedback TO lowercase
        REMOVE common punctuation
        SPLIT feedback INTO words
        positive_count = 0
        negative_count = 0
        FOR EACH word IN words DO
            IF word IN positive_words THEN
                positive_count = positive_count + 1
            END IF
            IF word IN negative_words THEN
                negative_count = negative_count + 1
            END IF
```

```
        END FOR
        score = positive_count - negative_count
        IF score > 0 THEN
            sentiment = "Positive"
        ELSE IF score < 0 THEN
            sentiment = "Negative"
        ELSE
            sentiment = "Neutral"
        END IF
        OUTPUT feedback
        OUTPUT positive_count
        OUTPUT negative_count
        OUTPUT score
        OUTPUT sentiment
    END FOR
END
```

Python Program:

```
positive_words = ["good", "excellent", "helpful", "clear", "interesting"]
negative_words = ["bad", "poor", "boring", "difficult", "confusing"]

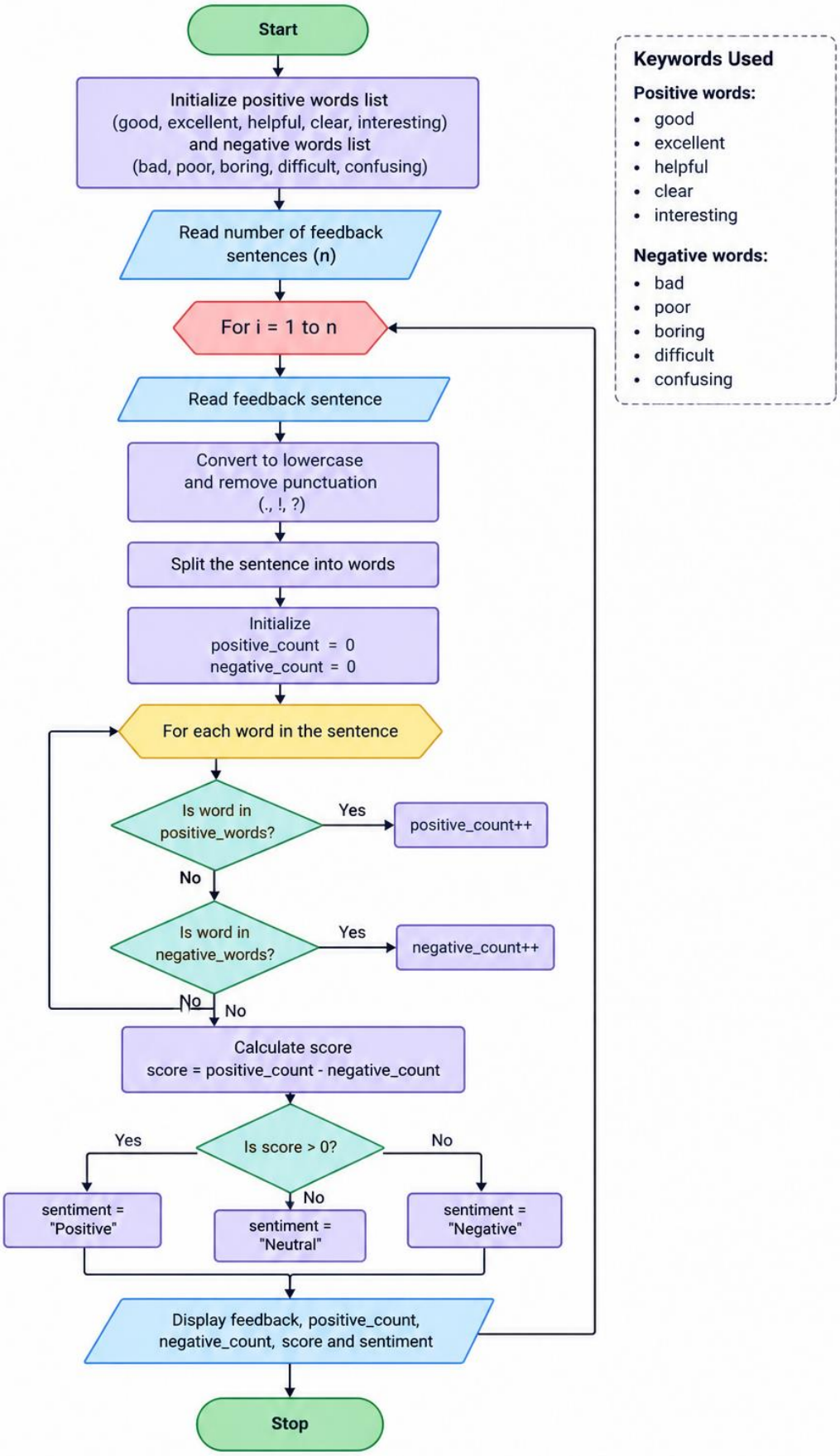
print("STUDENT FEEDBACK ANALYSIS")
print("-" * 55)

n = int(input("Enter the number of feedback sentences: "))

for i in range(n):
    feedback = input("\nEnter feedback: ")
    sentence = feedback.lower()
    for symbol in ".,!?":
        sentence = sentence.replace(symbol, "")
```

```
    words = sentence.split()
    positive_count = 0
    negative_count = 0
    for word in words:
        if word in positive_words:
            positive_count += 1
        if word in negative_words:
            negative_count += 1
    score = positive_count - negative_count
    if score > 0:
        sentiment = "Positive"
    elif score < 0:
        sentiment = "Negative"
    else:
        sentiment = "Neutral"
    print("\nFeedback:", feedback)
    print("Positive indicators:", positive_count)
    print("Negative indicators:", negative_count)
    print("Sentiment score:", score)
    print("Classification:", sentiment)
```

FlowChart:


Start
Initialize positive words list
(good, excellent, helpful, clear, interesting)
and negative words list
(bad, poor, boring, difficult, confusing)
Read number of feedback
sentences (n)
For i = 1 to n
Read feedback sentence
Convert to lowercase
and remove punctuation
(., !, ?)
Split the sentence into words
Initialize
positive_count = 0
negative_count = 0
For each word in the sentence
Is word in
positive_words?
Yes
positive_count++
No
Is word in
negative_words?
Yes
negative_count++
No
No
Calculate score
score = positive_count - negative_count
Is score > 0?
Yes
No
No
sentiment =
"Positive"
sentiment =
"Neutral"
sentiment =
"Negative"
Display feedback, positive_count,
negative_count, score and sentiment
Stop
Keywords Used
Positive words:
good
excellent
helpful
clear
interesting
Negative words:
bad
poor
boring
difficult
confusing

Output Screen Shots:

